



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0510 – COMPUTER VISION

EXPERIMENT NO: 14

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14. Perform transformation using Homography matrix.

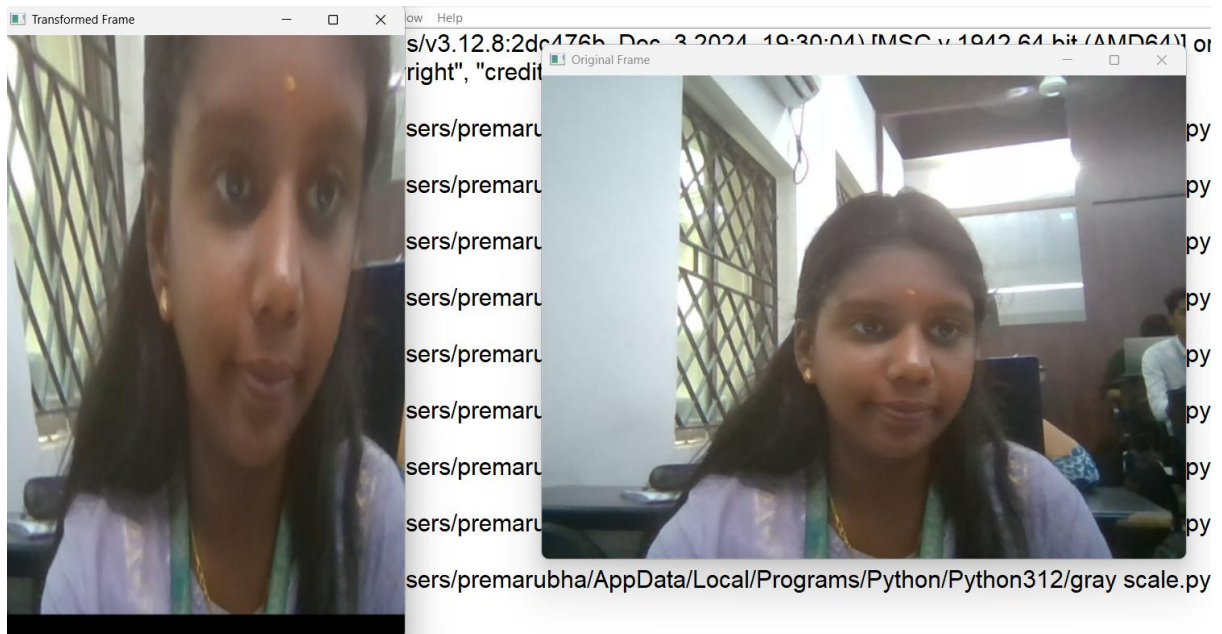
AIM:

Transformation using Homography matrix

PROGRAM:

```
import cv2
import numpy as np
roi_points = np.array([[150, 200], [450, 200], [550, 500], [50, 500]], dtype=np.float32)
target_points = np.array([[0, 0], [400, 0], [400, 600], [0, 600]], dtype=np.float32)
M = cv2.getPerspectiveTransform(roi_points, target_points)
cap = cv2.VideoCapture(0)
while True:
    ret, frame = cap.read()
    if not ret:
        break
    dst = cv2.warpPerspective(frame, M, (400, 600))
    cv2.imshow("Original Frame", frame)
    cv2.imshow("Transformed Frame", dst)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```

OUTPUT:



RESULT:

Thus the program is executed successfully.